Project Interim Report

On

File Compression Simulator
(Using Core Java & OOP Concepts)

RU-100-01-00012

Object Oriented Programming with Java

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Abstract

The Geometric Area Shape Calculator is a Java-based application designed to calculate the area of different geometric shapes such as circles, rectangles, triangles, squares, and other common figures. The project provides a simple menu-driven interface where users can select a shape, enter the required dimensions, and instantly obtain the calculated area.

This project is needed because manual area calculations can be time-consuming and prone to errors, especially when dealing with multiple shapes and formulas. By automating the calculation process, the application improves accuracy, saves time, and makes mathematical computations easier for students and general users.

The calculator uses Java programming concepts and Object-Oriented Programming (OOP) principles such as classes, objects, encapsulation, and method implementation. The system accepts user input, applies the appropriate geometric formula, and displays the result in a user-friendly format. The project serves as a practical example of applying Java programming skills to solve real-world mathematical problems while maintaining code organization and reusability.

Introduction

The Geometric Area Shape Calculator is a console-based Java application developed to calculate the area of various geometric shapes. Mathematics and geometry are important subjects in education and engineering, where area calculations are frequently required. This project simplifies these calculations by allowing users to choose a shape and enter the required measurements.

The application is designed using Object-Oriented Programming concepts, making the code modular, reusable, and easy to maintain. Different classes can be used to represent different geometric shapes, each containing methods for area calculation.

Features

Calculate area of a Circle

Calculate area of a Rectangle

Calculate area of a Triangle

Calculate area of a Square

Menu-driven interface

User-friendly interaction

Accurate mathematical calculations

OOP-based implementation

Scope

This project is suitable for students, teachers, and anyone needing quick area calculations for geometric shapes. It can be used as a learning tool for geometry and Java programming.

The project can be extended in the future by:

Adding volume calculations

Integrating a database to store previous calculations

Developing a graphical user interface (GUI)

Supporting more geometric shapes

Generating calculation reports

System Requirements

Hardware Requirements

Computer/Laptop
Minimum 4 GB RAM
Intel i3 Processor or higher
Keyboard and Monitor

Software Requirements

Java Development Kit (JDK)
VS Code / Eclipse / Edit Plus
Windows, Linux, or macOS Operating System

Methodology

The project follows a simple step-by-step process:
The program starts.
A menu displaying available geometric shapes is shown.
The user selects a shape.
The program asks for the required dimensions.
Radius for Circle
Length and Width for Rectangle
Base and Height for Triangle
Side for Square
The corresponding area formula is applied.
The calculated area is displayed.
The user can perform another calculation or exit the program.
The program terminates when the user chooses Exit.

Flowchart

```
START
  |
  V
Display Menu
  |
  V
Select Shape
  |
  V
Enter Dimensions
  |
  V
Calculate Area
  |
  V
Display Result
  |
  V
Another Calculation?
  |
 Yes ---------> Back to Menu
```

```
 |
No
 |
 V
END
```

Implementation

The project is implemented using Java classes and methods.
Main Class
The main class controls the execution of the program. It displays the menu, accepts user input, and calls the appropriate area calculation methods.
Circle Class
Stores the radius value and calculates the area using the formula:
Area = $\pi \times r^2$
Rectangle Class
Stores length and width values and calculates area using:
Area = Length × Width
Triangle Class
Stores base and height values and calculates area using:
Area = ½ × Base × Height
Square Class
Stores side length and calculates area using:
Area = Side × Side
The program uses conditional statements and methods to perform calculations based on the user's choice.

```
===== Geometric Area Shape Calculator =====

1. Circle
2. Rectangle
3. Triangle
4. Square
5. Exit

Enter Choice: 1

Enter Radius: 7

Area of Circle = 153.94

===== Geometric Area Shape Calculator =====

1. Circle
2. Rectangle
3. Triangle
4. Square
5. Exit

Enter Choice: 2

Enter Length: 10
Enter Width: 5

Area of Rectangle = 50
```

===== Geometric Area Shape Calculator =====

1. Circle
2. Rectangle
3. Triangle
4. Square
5. Exit

Enter Choice: 4

Enter Side: 8

Area of Square = 6

Planning & Structure

A Gantt Chart helps break the project into smaller and manageable tasks such as requirement analysis, system design, coding, testing, and documentation. Proper planning ensures that every stage of development is completed systematically.
Even though the project is relatively simple, creating a structured plan helps organize the development process and ensures smooth implementation. The project was designed using Object-Oriented Programming concepts, making the code modular and easier to maintain.

Time Management

The Gantt Chart helps in effective time management by showing:
- **When each task starts**
- **When each task should be completed**
- **Dependencies between tasks**
- **Overall project progress**

This approach helps avoid delays, reduces confusion, and ensures timely completion of the project without last-minute pressure.

Conclusion

The Geometric Area Shape Calculator project successfully demonstrates the implementation of Java programming and Object-Oriented Programming (OOP) concepts. The application provides an efficient and user-friendly way to calculate the area of different geometric shapes such as circles, rectangles, triangles, and squares.
The project reduces manual calculation effort, improves accuracy, and enhances understanding of geometry formulas and Java development principles. Through the use of classes, methods, and modular design, the project achieves good code organization and maintainability.
Overall, the project serves as a practical example of applying Java OOP concepts to solve real-world mathematical problems and can be further enhanced with additional features in the future.

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